Monte Carlo Pricing and Risk of Options and Interest-Rate Caps

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Summary. I price a European call and an interest-rate cap by Monte Carlo, validate both against closed-form formulas (Black–Scholes and Bachelier), and use the simulations to ask two risk questions: what the option means for the *client* who buys it, and what it means for the *bank* that sells it.

1. European call: pricing, then real outcomes

Inputs: $S = 100$, $K = 102$, $T = 3$, $r = 10\%$, $\sigma = 20\%$. For pricing, the stock grows at the risk-free rate (risk-neutral measure):

$$S_T = S\exp\big[(r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z\big], \qquad \hat{C} = \frac{1}{N}\sum_{i=1}^{N} e^{-rT}\max(S_T^{(i)} - K, 0), \qquad \mathrm{SE} = \frac{s}{\sqrt{N}}.$$

Validation. The Black–Scholes price is $C = 27.97$. From 10^3 to 10^6 paths, $|\hat{C} - C|/\mathrm{SE} < 2$ throughout, and the standard error falls by $\approx \sqrt{10}$ per tenfold increase in paths, so the simulation agrees with the formula.

Real outcomes. The price does not depend on the stock's expected return μ, but real outcomes do. Re-simulating with drift μ (still discounting at r), the buyer's result as a share of the premium, relative to a risk-free account, is:

	$\mu = 5\%$	Risk-neutral ($r = 10\%$)	$\mu = 15\%$	$\mu = 20\%$
Lose the whole premium	42%	26%	14%	7%
Beat the risk-free account	24%	39%	56%	72%
Median return	−80%	−34%	+21%	+84%
Mean return	−39%	≈ 0%	+51%	+115%

The price (27.97) is the same in every column; only the outcomes change, and they depend on μ, which cannot be observed. The payoff is skewed: many small losses and a few large gains, so the **mean is well above the median**. In the risk-neutral column the mean is zero (a fair price), yet the typical buyer loses a third of the premium.

2. Delta hedging and model risk

The bank sells the call and holds $\Delta = N(d_1)$ shares, rebalancing over time. Its per-option P&L (negative = loss), with $\mu = 15\%$, priced and hedged at $\sigma = 20\%$:

Seller's position	Average P&L	99% VaR	99% ES
Unhedged	−14.6	−142.8	−175.9
Delta-hedged, weekly	≈ 0	−1.6	−2.1
Delta-hedged, daily	≈ 0	−0.8	−1.0
Hedged weekly, real volatility 30%	−4.4	−11.7	−12.9

Hedging removes almost all the risk and makes the result independent of μ. But if the stock is more volatile than assumed, the hedged seller loses on almost every path:

$$d\Pi = \tfrac{1}{2}\,\Gamma S^2\big(\sigma^2 - \sigma_{\mathrm{real}}^2\big)\,dt < 0 \quad \text{when } \sigma_{\mathrm{real}} > \sigma,$$

so the remaining risk is *model risk*.

3. Interest-rate cap: Bachelier model

A company borrows £10m for 3 years at a quarterly-resetting rate and buys a 4% cap (11 caplets). Rates can go negative, so each future rate is modelled as normal, $L_i \sim N(F, \sigma_N^2 T_i)$, with $F = 4\%$ and $\sigma_N = 0.8\%$:

$$\mathrm{caplet}_i = A\,\tau\,D_i\Big[(F - K)N(d_i) + \sigma_N\sqrt{T_i}\,n(d_i)\Big], \qquad d_i = \frac{F - K}{\sigma_N\sqrt{T_i}}.$$

The cap price is £95,049; a Monte Carlo of 100,000 rate paths gives £95,041 ($|\text{error}|/\text{SE} \approx 0.02$). For the bank selling it unhedged, the 99% VaR is about $-$£386,000, roughly four times the premium.

Takeaways

- When advising a client, I would use the median rather than the mean, because it describes what typically happens to someone who buys one option. The mean is pushed up by a few very large gains. I would also show the results for a range of growth rates, since we cannot know the real one, and be clear about when the client is better off keeping the money in a risk-free account: if the stock grows at 5%, three out of four buyers do worse than the account, and two out of five lose everything.
- For the bank, hedging is what matters. Delta hedging removes almost all the risk of selling the option, but it relies on the volatility being right. When the stock moved more than assumed, the hedged position lost money on almost every path.
- The cap example shows that Black–Scholes is not always enough. It assumes prices can never go below zero, which failed when interest rates turned negative. A model that allows for this, such as Bachelier, is needed to describe what can actually happen.